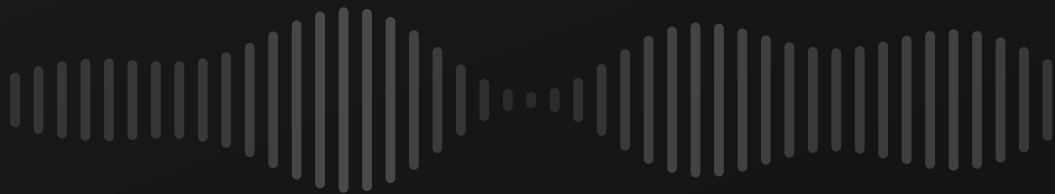




# Audian

## User & Technical Manual

Voice dictation for Windows — speak naturally and get polished text wherever you type. Private by default: everything runs on your PC.



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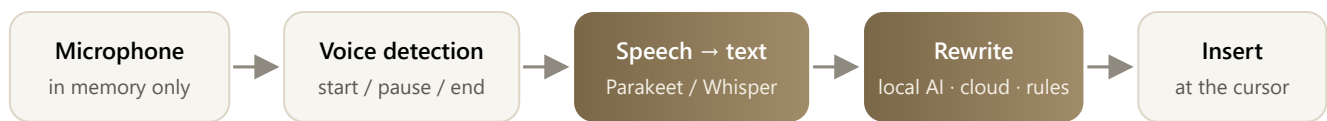
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# 1 What Audian is

Audian is a small background utility. You press a keyboard shortcut in any app, speak the way you talk, and Audian inserts a clean, well-formed version of what you *meant* — not a literal transcript — at your text cursor.

Saying *"um so basically I, I want to make this thing work better and like organize the code because it's kinda messy"* produces **"I want to make this work better and organize the code, because it's kind of messy."** in Natural mode, or **"Improve the implementation and organize the code to make it cleaner and more maintainable."** in AI Prompt mode.

## The pipeline



Typically 0.1 – 1.5 s from the end of your speech to text on screen

## Design principles

- **Private by default.** Speech recognition and rewriting run on your PC: rewriting on your graphics card if you have a dedicated one, otherwise on the CPU. Nothing is sent anywhere unless you choose the Antigravity cloud provider.
- **Invisible until needed.** The tray app idles at ~11 MB of RAM and 0 % CPU. The AI engines are separate processes that start when you press the shortcut and exit as soon as your text is inserted.
- **Works everywhere you type.** Text is pasted through the clipboard, which every Windows app understands — editors, browsers, chat apps, Office, terminals — and your clipboard is restored afterwards.
- **Yours to style.** Six colour themes — Ivory (off-white, the default), Violet, Ocean, Mint, Sunset and Rose — for the window, the logo and the recording indicator; a light or dark window (or one that follows Windows); and a dark or light indicator.
- **Fails gracefully.** If the AI fails, a basic cleanup is inserted; if the window closed or is elevated, the text goes to your clipboard and a notification explains why.

**11 MB**

tray app RAM while idle  
(0 % CPU)

**0.1–1.5 s**

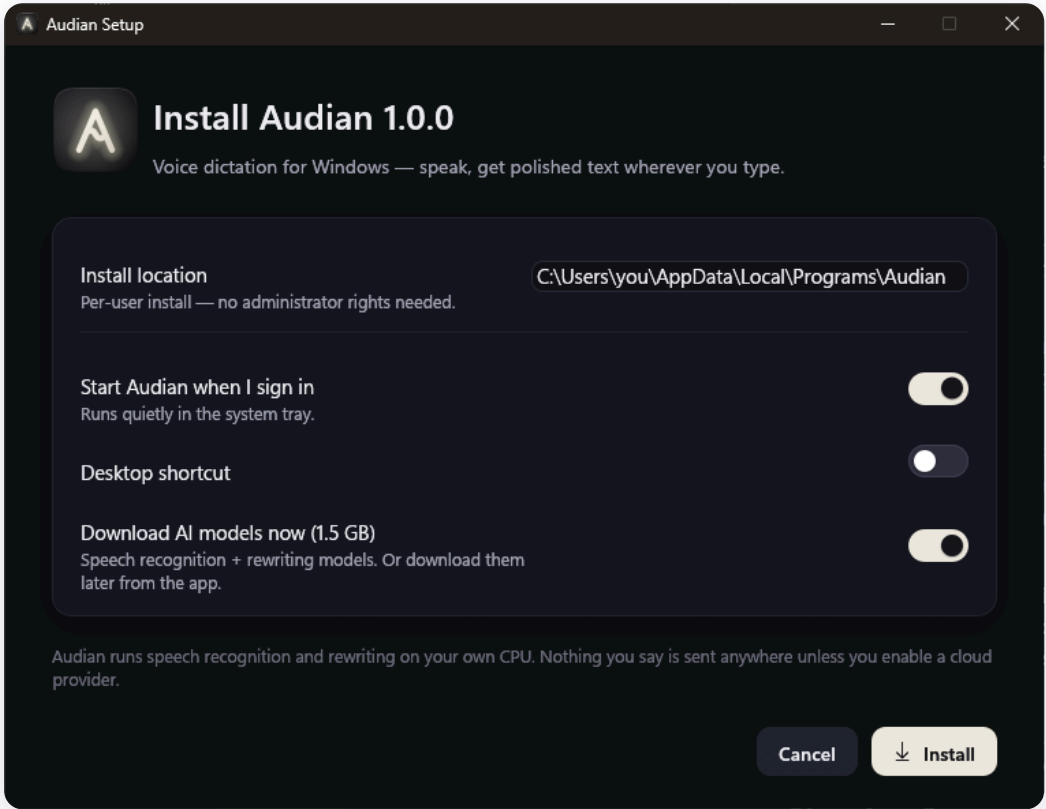
end of speech → text,  
with or without a graphics card

**0 MB**

video memory used between  
dictations — runs on any PC

## 2 Installing & first run

Run `Audian-Setup-1.2.0-x64.exe` . It installs for your Windows account only, so no administrator rights are needed (the same model as VS Code's user installer). Requirements: Windows 10 or 11, 64-bit x86 CPU with AVX2 (any Intel/AMD CPU from roughly 2015 on), 8 GB RAM recommended (4 GB works with the lighter models). A graphics card is optional: with a dedicated one, rewriting runs on it.



The installer. Options: location, start with Windows, desktop shortcut, and downloading the AI models (about 2 GB) during setup.

### What the installer does

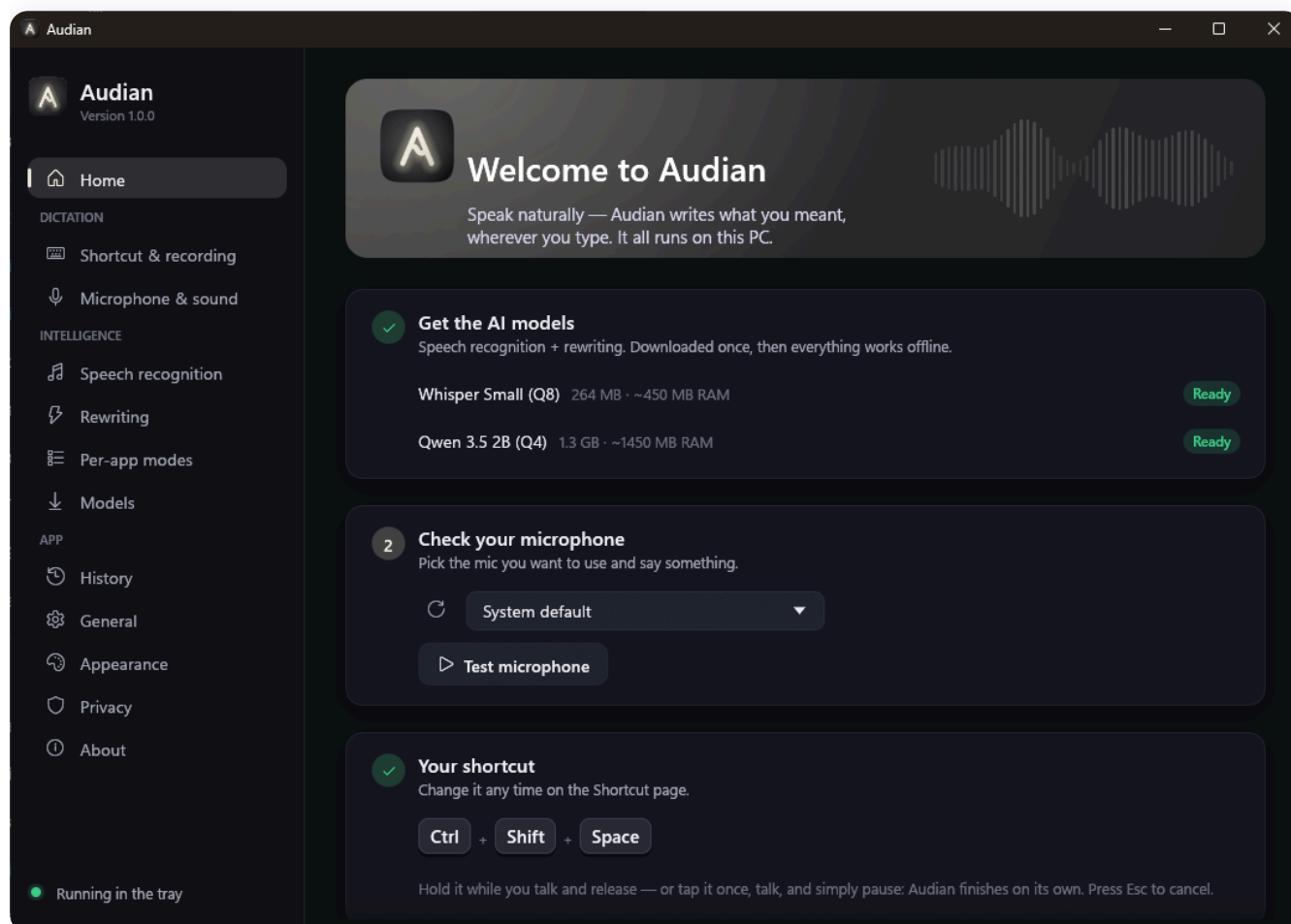
| Step         | Details                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Files        | <code>%LOCALAPPDATA%\Programs\Audian</code> : <code>audian.exe</code> (tray app + window), <code>audian-stt.exe</code> (Whisper speech engine), <code>audian-parakeet.exe</code> (Parakeet speech engine), <code>audian-llm.exe</code> (rewriting engine), <code>audian-llm-gpu.exe</code> (the same engine for graphics cards), <code>uninstall.exe</code> . About 100 MB. |
| Shortcuts    | Start Menu entry "Audian"; optional desktop shortcut.                                                                                                                                                                                                                                                                                                                       |
| Registration | Listed in <i>Settings &gt; Apps &gt; Installed apps</i> with a working Uninstall button.                                                                                                                                                                                                                                                                                    |
| Startup      | Optional entry in your user's <i>Run</i> key (Audian starts in the tray when you sign in).                                                                                                                                                                                                                                                                                  |
| Models       | Optional download of NVIDIA Parakeet (671 MB) and Qwen 3.5 2B (1.28 GB) into <code>%LOCALAPPDATA%\Audian\models</code> , verified by SHA-256. Resumable. Whisper can be added later from the Models page.                                                                                                                                                                   |
| Updates      | Running a newer installer stops Audian, replaces the files and keeps your settings, history and models.                                                                                                                                                                                                                                                                     |

## Silent install (for scripts, winget, etc.)

```
Audian-Setup.exe --quiet [--with-models] [--no-autostart] [--desktop-shortcut] [--no-launch]
uninstall.exe --uninstall [--quiet] [--purge]      # --purge also deletes settings, history, models
```

## First run

On first launch Audian opens a short welcome: (1) download the models if they're missing, (2) pick and test your microphone, (3) learn the shortcut. After that it lives in the system tray.



The first-run welcome.

### 3 Dictating

Click into any text field, then use your shortcut (default `Ctrl` + `Shift` + `Space`; yours is set to `Ctrl` + `Shift` + `V`).

| Style                      | How it works                                                                                                                                   | Best for                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Hybrid</b><br>(default) | Hold the shortcut while speaking and release to finish — or tap it once and just talk; Audian finishes when you pause (or when you tap again). | Everything                 |
| <b>Push-to-talk</b>        | Records only while the shortcut is held.                                                                                                       | Short phrases, noisy rooms |
| <b>Toggle</b>              | Press to start; press again or pause to finish.                                                                                                | Long, hands-free dictation |

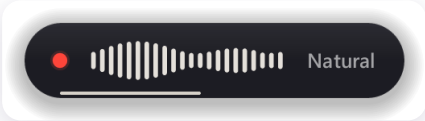
`Esc` cancels a dictation at any point. `Ctrl` + `Shift` + `Alt` + `V` pastes your last dictation again (configurable).

#### The recording indicator

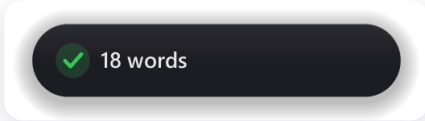
A small pill appears near your text cursor (or at the bottom of the screen if the app doesn't expose its cursor). It never takes focus and clicks pass through it.



**Listening** — live waveform; label shows the mode. A ring around the dot means hands-free.



**Pause detected** — a line in your theme's colour fills as auto-stop approaches; speaking again resets it.



**Done** — word count, then the pill fades. Errors show an amber mark and a short reason.

While processing, the pill shows *Transcribing...* then *Polishing...*. Soft chimes mark start and stop (can be turned off under *Microphone & sound*). Its colours, dark or light style, position and label are set under *Appearance*.

#### Where the text goes

- Audian remembers the window you were in when you pressed the shortcut, and pastes there — even if you clicked elsewhere while it was processing (it switches back first).
- Your clipboard is saved before pasting and restored ~0.6 s later. Dictated text is excluded from `Win` + `V` clipboard history and cloud clipboard.
- In terminals, multi-line results are joined into one line so nothing runs by accident. mintty/PuTTY receive `Shift` + `Insert`.
- If the window was closed, or runs as administrator (Windows blocks input from normal apps into elevated ones), the text is placed on the clipboard and a notification tells you to press `Ctrl` + `V`.

## 4 Knowing when you're done

In hands-free dictation Audian decides when you've finished speaking — quickly when you've completed a sentence, patiently when you pause mid-thought.

### How it decides

1. **Voice activity detection** runs on every 10 ms of audio and tracks your room's background noise, so it knows the difference between “quiet” and “speaking” on any microphone.
2. When you pause ( $\approx 0.5$  s), Audian quietly **transcribes what you've said so far** (a “preview”).
3. If the preview sounds **finished** (“...send it to Maria.”), Audian stops after your *pause tolerance* (default 2.0 s).
4. If it sounds **unfinished** — it ends with “and”, “but”, “because”, “the”, a comma, “um”, or Portuguese “e”, “mas”, “porque”, “que”, “de”... — Audian waits the *extra patience* as well (default +1.5 s).
5. When it stops, the preview is **reused**. If it sounded finished, the local AI has **already rewritten it** during the pause, so the text appears the moment Audian stops listening; otherwise it appears 0.1–1 s later.

#### Tuning it

Shortcut & recording › Finish automatically. Presets: **Quick** 1.2 s, **Balanced** 2 s, **Patient** 3 s, or any value 0.6–6 s. You can also disable smart detection (pure silence timer) or auto-stop entirely (then tap the shortcut to finish). If nothing is said for 8 s, the dictation is cancelled.

### Measured behaviour (test recordings)

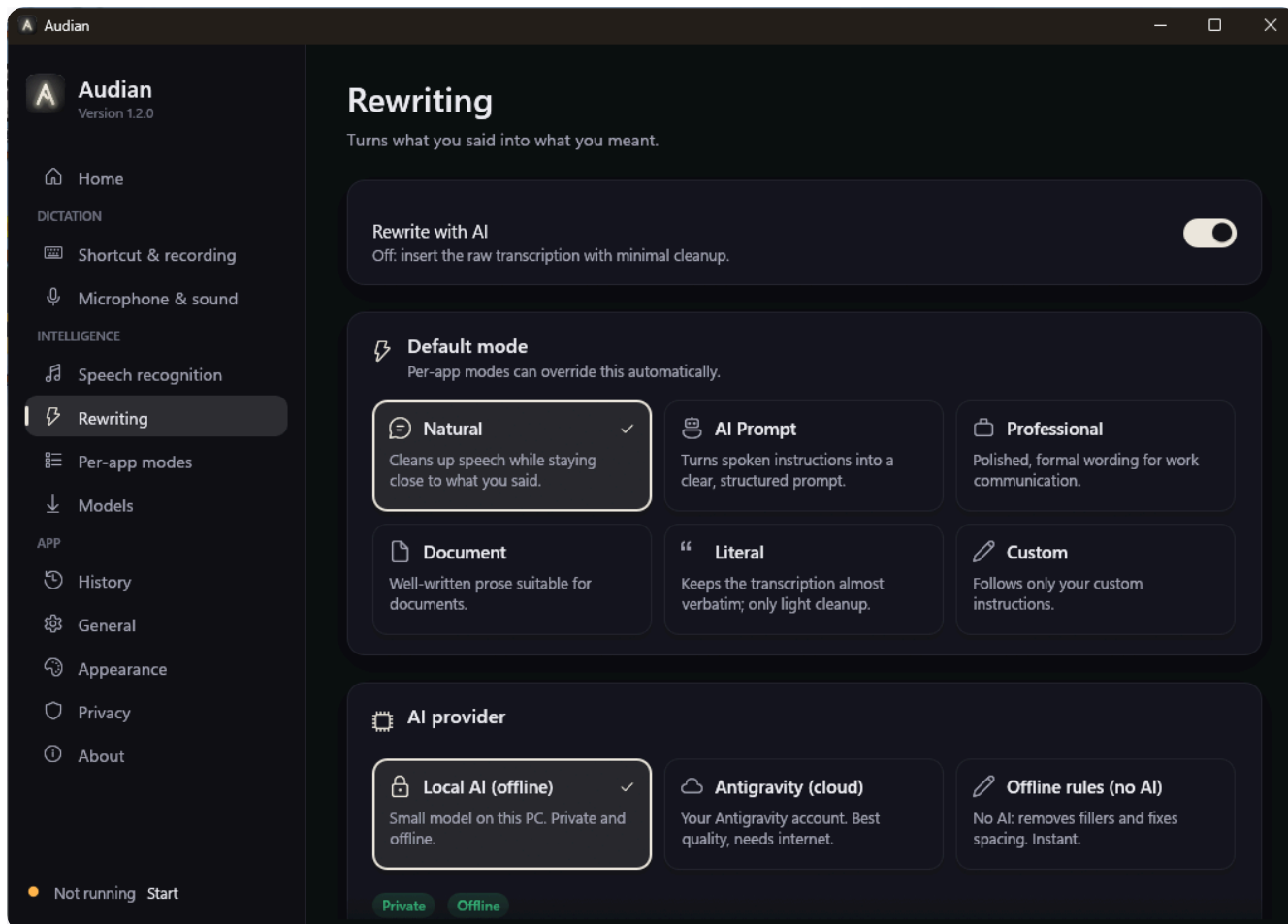
| What was said                                                                                           | Result                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| “Let's meet on Wednesday at three.” → silence                                                           | Stopped 1.5–2.0 s after the sentence; text appeared 0.98 s later (transcript reused).                                                           |
| “I think we should push the release and” → 2.2 s pause → “also update the documentation before Friday.” | Did <b>not</b> stop at the pause (sentence unfinished); captured everything; stopped 2.07 s after the end; text 0.69 s later.                   |
| “Send the report to Maria.” → 2.2 s pause → “Also call John.”                                           | Stopped after the first sentence (complete + pause longer than the tolerance). Choose <i>Patient</i> if you take long breaks between sentences. |

## 5 Rewriting modes & providers

### Modes

| Mode                | Purpose                                              | Example output (from real tests)                                                                                                                                                                                      |
|---------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Natural</b>      | Clean speech, stay close to your words.              | "Let's meet on Wednesday at 3. Can you send the report to Maria before that?" (from "Let's meet on Tuesday — no, wait, Wednesday...")                                                                                 |
| <b>AI Prompt</b>    | A clear, structured instruction for an AI assistant. | "Add a settings page where users can select their microphone and choose a shortcut. Include a toggle to enable or disable Windows startup, and ensure all settings are saved to a config file and loaded on startup." |
| <b>Professional</b> | Polished wording for work email/chat.                | "For the new feature, please add a settings page where users can select their microphone, configure a shortcut, and toggle launching on Windows startup..."                                                           |
| <b>Document</b>     | Prose for documents, with paragraphs.                | —                                                                                                                                                                                                                     |
| <b>Literal</b>      | Almost verbatim; no AI.                              | Whisper's transcript with recognizer artefacts removed.                                                                                                                                                               |
| <b>Custom</b>       | Only your custom instructions.                       | e.g. "Always organize my spoken instructions into concise technical prompts."                                                                                                                                         |

In every mode the rewriter **never answers** what you say: "uh what's the, what's the capital of France" becomes "What's the capital of France?", not "Paris". It keeps your language (Portuguese stays Portuguese), your point of view, names, numbers and code identifiers.



Rewriting: default mode, AI provider and options.

## Self-corrections

People correct themselves while speaking: “*Vamos marcar para as 9 da noite. Não, na verdade, eu errei, é às 8*”, “*Send it to Mark, I mean to Sarah*”. Audian keeps only the corrected version (“...às 8 da noite”, “Send it to Sarah”). Small models easily miss a correction in the middle of a sentence, so before rewriting Audian looks for correction phrases — *na verdade, quer dizer, ou melhor, digo, aliás, me enganei, corrigindo, desculpa, I mean, actually, or rather, scratch that, make that, no wait, sorry*, Spanish and French equivalents, and a “no”/“não” between commas — and points the model at that passage. Phrases at the very start (“Na verdade, eu gosto...”, “Actually, I like...”) are left alone, and the model still decides whether it really is a correction.

| Test set (30 dictated sentences)                      | 1.1.0 | 1.2.0 |
|-------------------------------------------------------|-------|-------|
| Portuguese self-corrections resolved (13)             | 4     | 11–13 |
| English self-corrections resolved (9)                 | ~5    | 9     |
| Sentences without a correction changed by mistake (5) | 0     | 0     |

Both columns use Qwen 3.5 2B (*Balanced*); the range is the graphics card and the CPU, whose tiny numeric differences tip a few borderline sentences either way. The 0.8B model (*Fast*) resolved 0 of the 13 Portuguese and 1 of the 9 English corrections in the same test, and shortened two sentences in ways that changed them.

## Providers

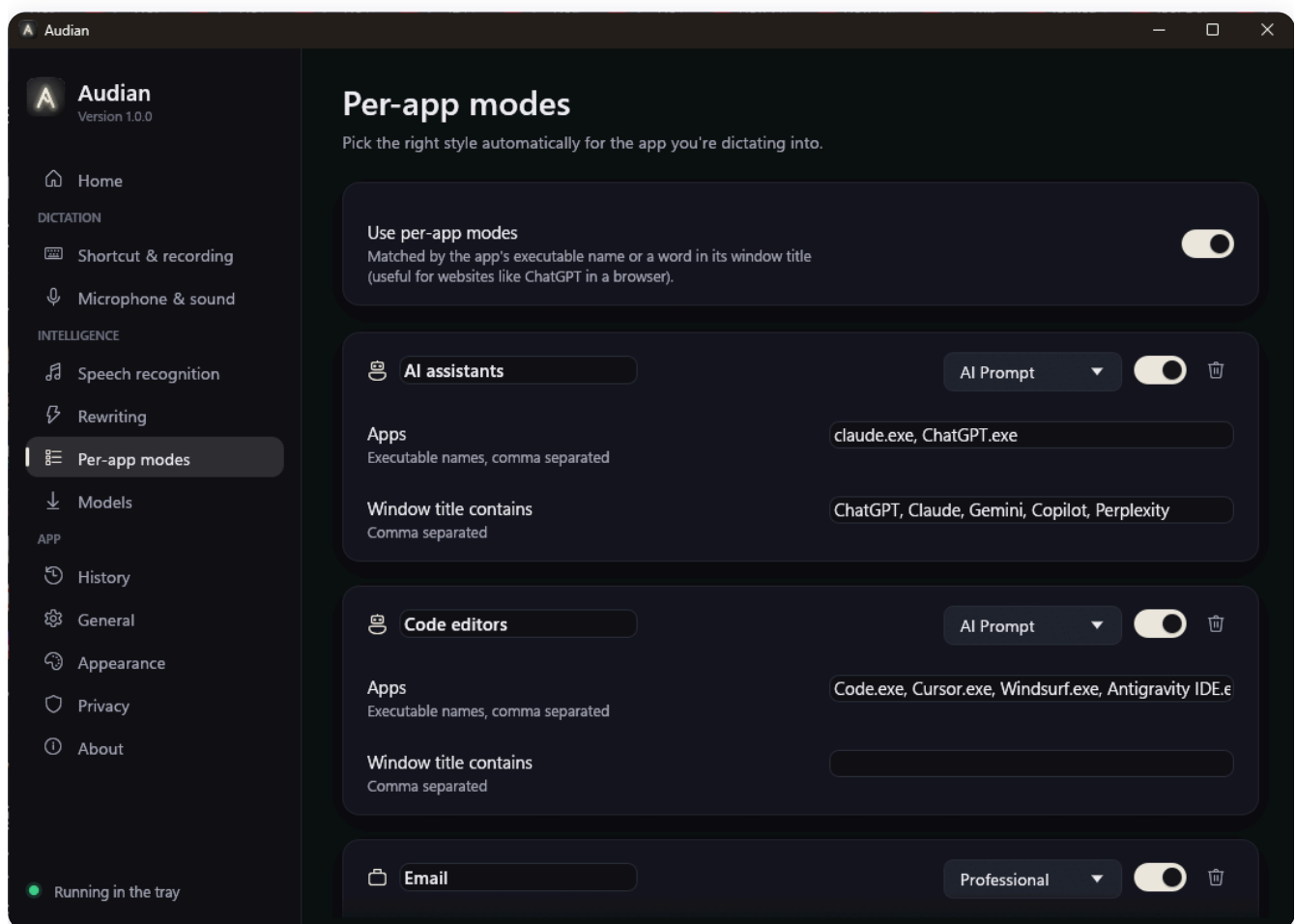
| Provider                           | Where text goes                                 | Speed                                     | Notes                                                                                                                                          |
|------------------------------------|-------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Local AI</b><br><b>default</b>  | Nowhere — runs on your graphics card or CPU     | ≈0.1 s (graphics card)<br>0.5–1.5 s (CPU) | Qwen 3.5 2B ( <i>Balanced</i> ) or 0.8B ( <i>Fast</i> ); private and offline.                                                                  |
| <b>Antigravity</b><br><b>cloud</b> | Your Antigravity account (Gemini/Claude models) | 4–8 s                                     | Best on long, rambling dictation. Audio never leaves the PC — only text. Each dictation appears as a conversation in your Antigravity history. |
| <b>Offline rules</b>               | Nowhere                                         | instant                                   | Removes “um/uh”, stutters (“I I”), fixes spacing/capitals. Also the automatic fallback when an AI provider fails.                              |

## 6 Per-app modes & translation

### Per-app modes

Audian checks which app you're dictating into and picks the mode for it automatically. Rules match the program's executable name or a word in its window title (so "ChatGPT" in a browser tab works too).

| Profile (default) | Mode         | Matches                                                                                                 |
|-------------------|--------------|---------------------------------------------------------------------------------------------------------|
| AI assistants     | AI Prompt    | claude.exe, ChatGPT.exe; titles containing ChatGPT, Claude, Gemini, Copilot, Perplexity                 |
| Code editors      | AI Prompt    | Code.exe, Cursor.exe, Windsurf.exe, Antigravity IDE.exe, zed.exe, devenv.exe, idea64.exe, pycharm64.exe |
| Email             | Professional | OUTLOOK.EXE, olk.exe, thunderbird.exe; titles Gmail, Outlook                                            |
| Documents         | Document     | WINWORD.EXE, soffice.bin, Notion.exe, Obsidian.exe; title Google Docs                                   |
| Work chat (off)   | Professional | ms-teams.exe, Teams.exe, slack.exe                                                                      |



Per-app modes are fully editable; the overlay label shows which mode is active.

## Output language (translate as you dictate)

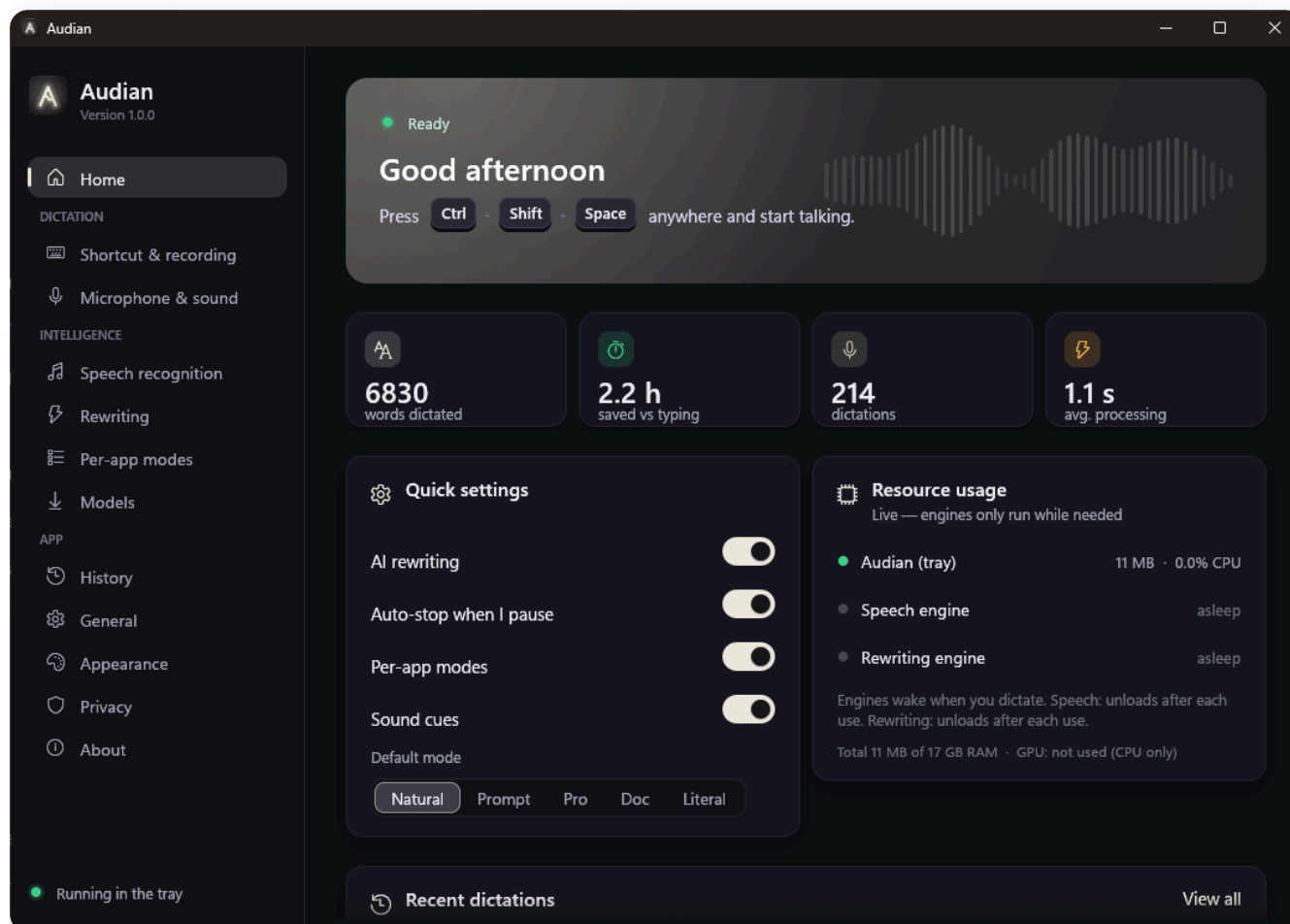
Under *Rewriting* › *Output language* (or the tray menu) choose a language to always write in. Speak Portuguese, get English — or the reverse. Tested results: “Você pode me explicar como funciona o borrow checker do Rust?” → “Can you explain how the Rust borrow checker works?”; “Can you check if the build is passing before we merge?” → “Pode verificar se o build está passando antes de fazer o merge?”. English and Portuguese have built-in examples for the local model; other target languages work best with the Antigravity provider.

## Recognizing two languages

With *Spoken language: Auto-detect*, tick the languages you actually speak (e.g. English + Portuguese). Detection then only chooses among those — avoiding mix-ups such as short Portuguese phrases being detected as Galician or Spanish — and is faster.

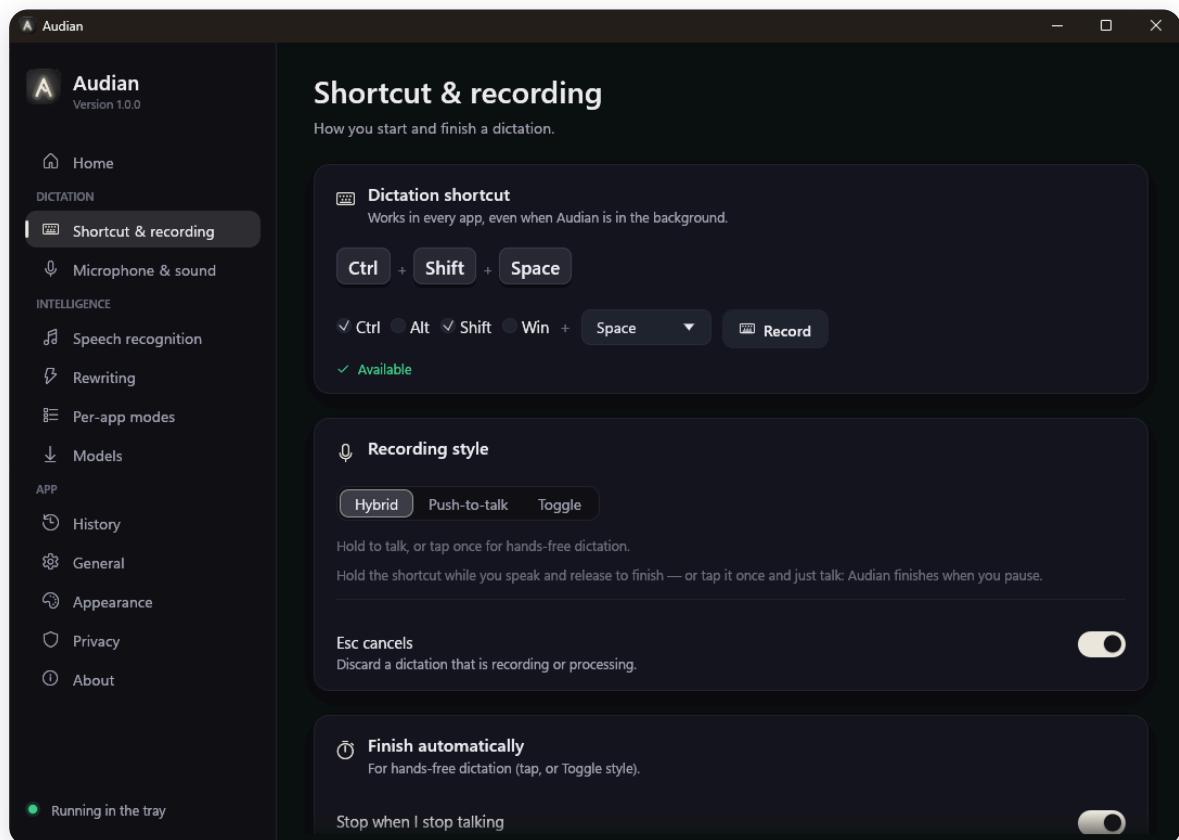
## 7 The Audian window

Left-click the tray icon to open it (right-click shows a quick menu with mode, provider, output language, microphone, auto-stop and per-app toggles). All changes are saved and applied instantly.

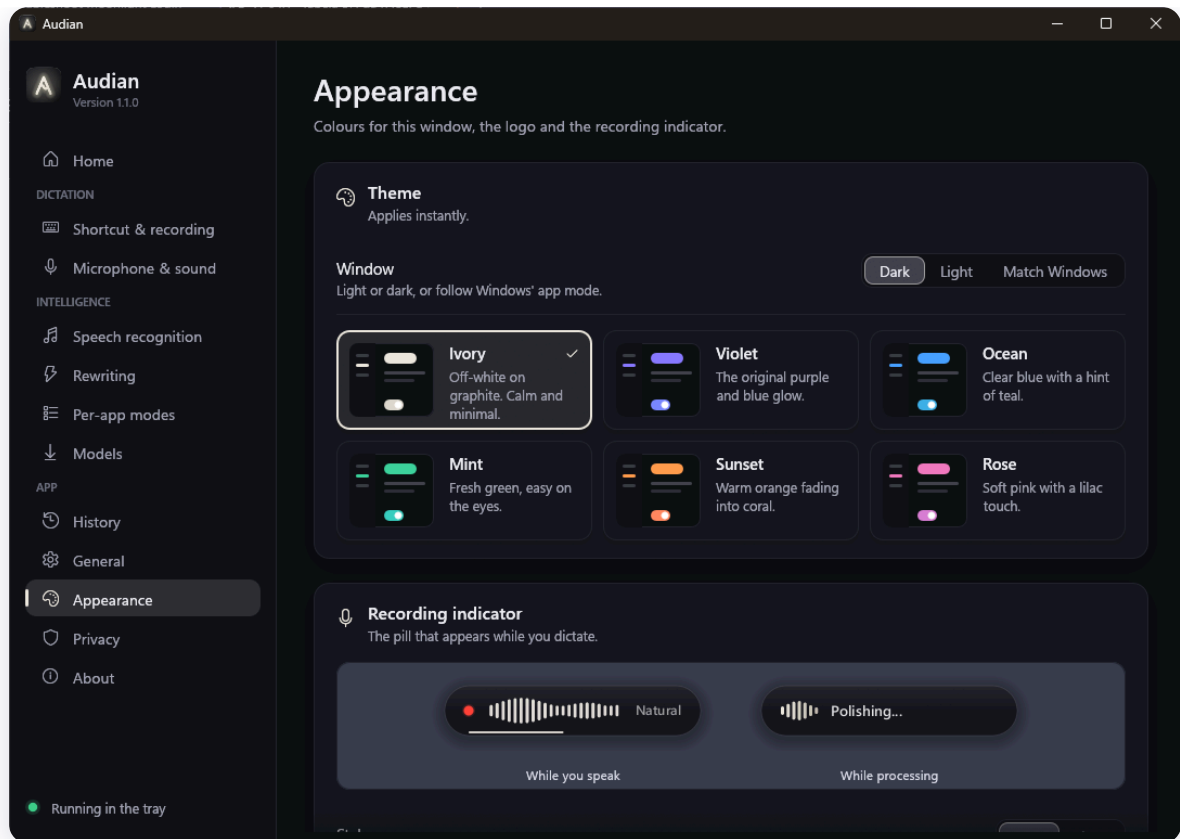


Home: status, your shortcut, statistics, quick settings, live resource usage and recent dictations.

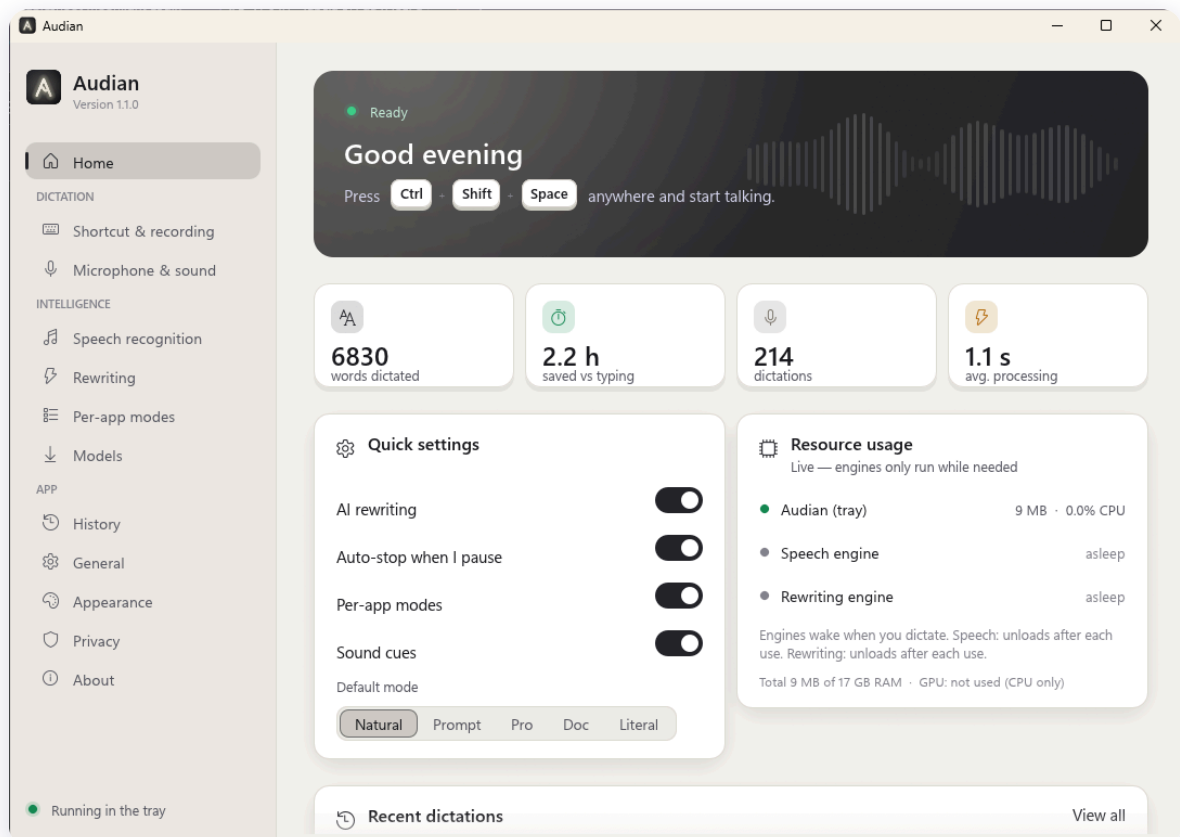
| Page                 | What you'll find                                                                                                                                     |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Home                 | Status, words dictated, time saved vs typing, quick toggles, live engine usage, recent dictations.                                                   |
| Shortcut & recording | Shortcut editor with conflict check, recording style, auto-stop tuning, Esc-to-cancel, paste-last shortcut.                                          |
| Microphone & sound   | Device, live level meter, gain, voice-detection sensitivity, silence threshold, max length, sound cues.                                              |
| Speech recognition   | Engine (NVIDIA Parakeet or Whisper), model, CPU threads, how long the model stays loaded, languages, custom vocabulary.                              |
| Rewriting            | AI on/off, default mode, provider, model ( <i>Balanced</i> or <i>Fast</i> ), graphics card on/off, output language, custom instructions, fallback.   |
| Per-app modes        | Automatic mode rules.                                                                                                                                |
| Models               | Download, switch and delete models; sizes, RAM and speed for each.                                                                                   |
| History              | Searchable recent dictations with what you originally said, timing and provider; copy or clear.                                                      |
| General              | Start with Windows, notifications, insertion method, clipboard restore, terminals.                                                                   |
| Appearance           | Light or dark window (or match Windows), colour theme (window, logo, indicator); recording indicator style, position and label, with a live preview. |
| Privacy              | Data-flow summary, history, recordings, transcript logging, statistics.                                                                              |
| About                | Version, live resource usage, file locations.                                                                                                        |



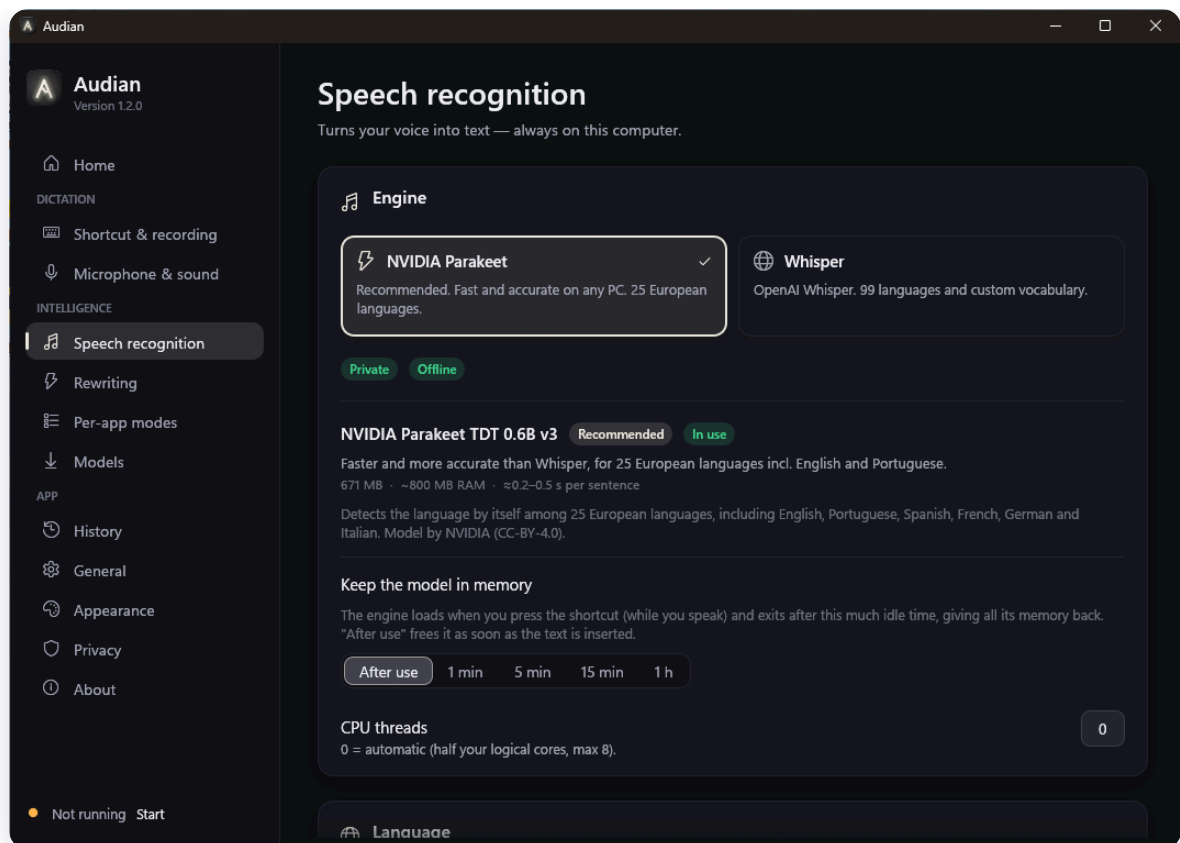
Shortcut & recording: the shortcut editor checks availability as you type.



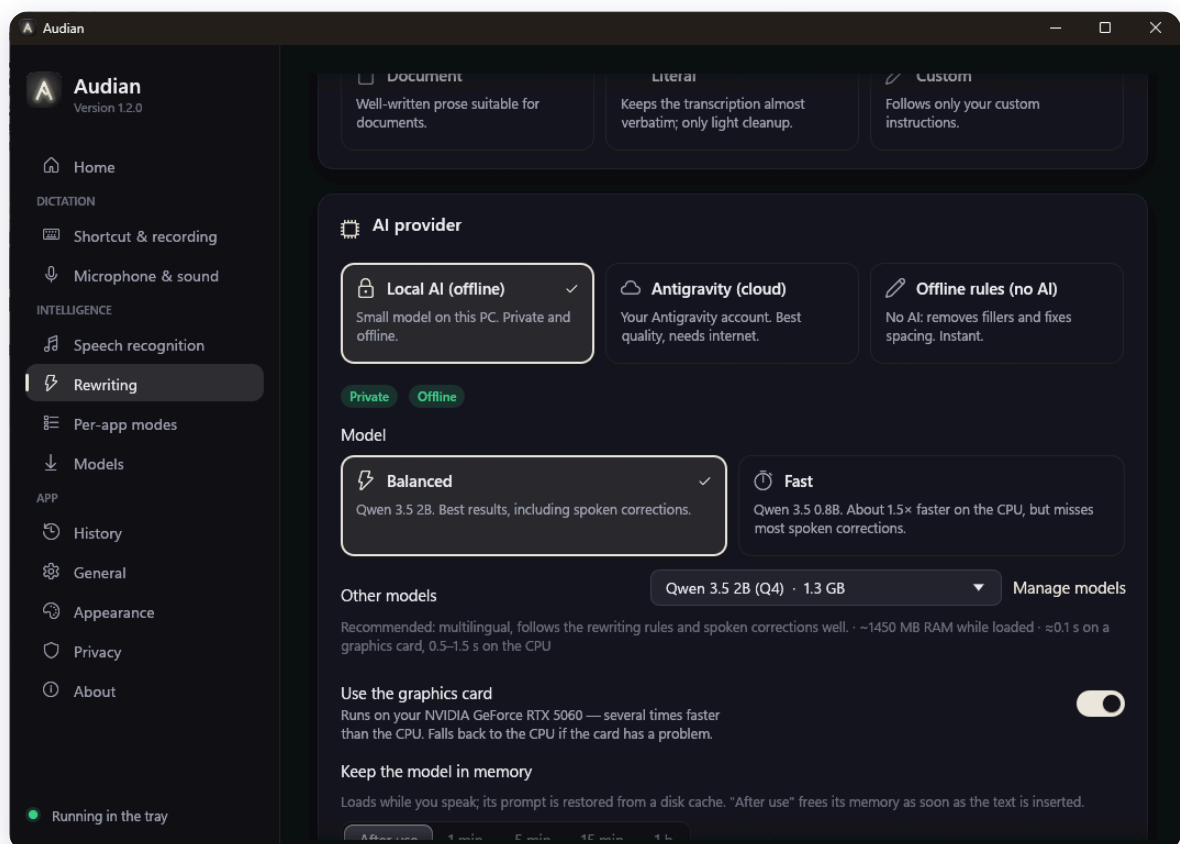
Appearance: light or dark window, six themes that apply instantly, and the recording indicator in dark or light style.



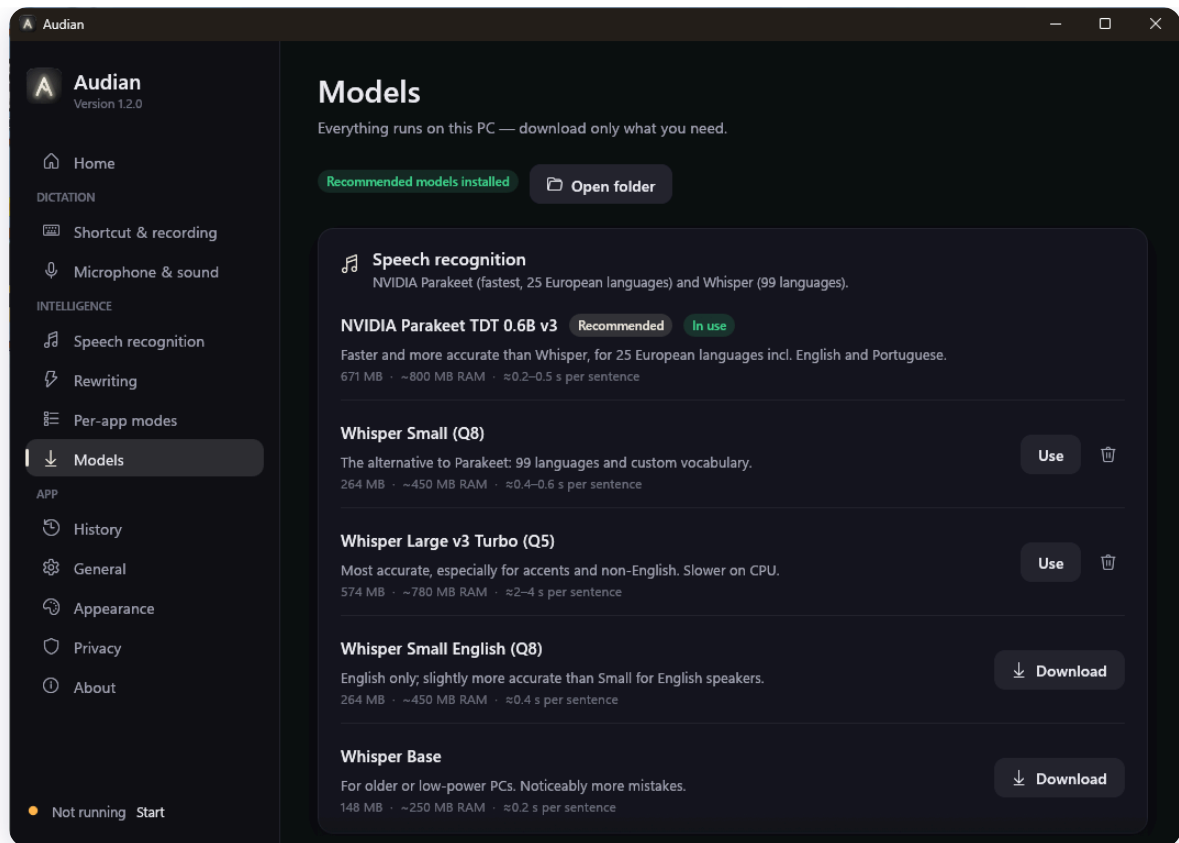
The window in light mode. Accents switch to each theme's darker shade so they stay readable.



Speech recognition: NVIDIA Parakeet (the default) or Whisper; the engine exits right after each dictation by default.



Rewriting with the local AI: the Balanced or Fast model, and the graphics card Audian found.



Models: download, switch and remove, with size, RAM and speed for each.

## 8 Resource usage (measured)

All numbers were measured on the development PC: Intel Core i9-12900KS (16 cores / 24 threads), 16 GB RAM, Windows 11, NVIDIA RTX 5060 (8 GB). Speech recognition always runs on the CPU; rewriting was measured on both the graphics card and the CPU.

### The tray app

| State                           | RAM (working set)                             | CPU                                                       |
|---------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| Idle (waiting for the shortcut) | <b>11 MB</b> (1.9 MB private)                 | <b>0 ms</b> in 30 s — no timers or polling run while idle |
| Recording (15.7 s dictation)    | ~38 MB (audio buffer, overlay)                | 0.42 s total = <b>0.11 %</b> of the machine               |
| Right after a dictation         | <b>1.5 MB</b> (memory handed back to Windows) | 0                                                         |
| Audian window open              | ~150 MB (separate process, OpenGL)            | only while animating                                      |

### AI engines (separate processes)

| Engine / model                                                                | Load time                                                 | Peak RAM                         | CPU while working                       | Speed                                           | VRAM   |
|-------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------|-----------------------------------------|-------------------------------------------------|--------|
| Speech: Whisper Small Q8                                                      | 0.45 s + 0.25 s warm-up (while you speak)                 | <b>~450 MB</b>                   | 8 threads (~7 cores, ~30 % of this CPU) | 6.6 s clip: 0.4–0.6 s<br>14.6 s clip: 1.0–1.4 s | 0      |
| Speech: Whisper Large v3 Turbo Q5                                             | 2.2 s                                                     | <b>777 MB</b>                    | ~7.6 cores                              | 14.6 s clip: 4.4–7.9 s                          | 0      |
| Speech: NVIDIA Parakeet TDT 0.6B v3 <b>default</b>                            | 1.5 s (while you speak)                                   | <b>~0.8 GB</b><br>0.7 GB private | 8 threads                               | 5.9 s clip: 0.2 s<br>14.6 s clip: 0.5 s         | 0      |
| Rewriting: Qwen 3.5 2B Q4 on the RTX 5060 <b>default with a graphics card</b> | 1.1–1.5 s (while you speak)                               | <b>0.6 GB</b>                    | under 1 core                            | 0.1–0.3 s per rewrite                           | 1.4 GB |
| Rewriting: Qwen 3.5 2B Q4 on the CPU                                          | 0.5 s + 25 ms prompt restore from cache (while you speak) | <b>1.4 GB</b><br>0.5 GB private  | ~5.4 cores                              | 0.5–1.5 s per rewrite (~30 tokens/s)            | 0      |
| Rewriting: Antigravity                                                        | CLI start ≈4 s (hidden while you speak)                   | not measured (CLI process)       | negligible (runs in the cloud)          | 4–8 s                                           | 0      |

By default both engines **exit as soon as your text is inserted** ("Keep the model in memory: After use"), so between dictations Audian uses only the tray app's ~11 MB. Because the engines start the moment you press the shortcut and load while you speak, this costs almost nothing: in a measured dictation starting from nothing

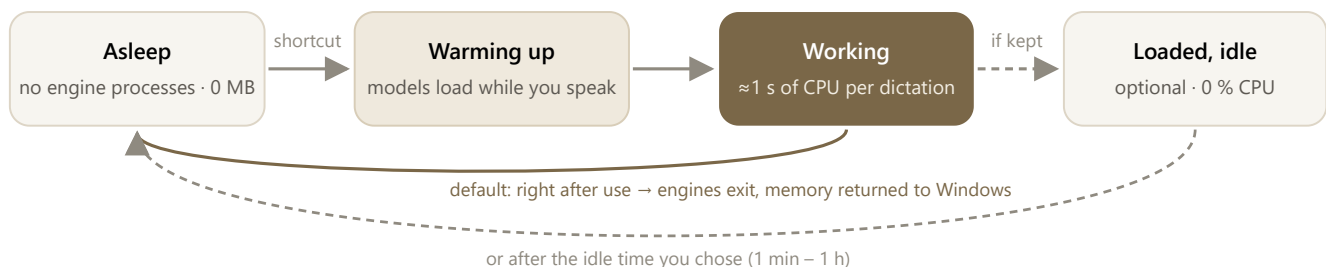
loaded, the text was inserted 1.9 s after the recording ended (0.6 s speech recognition + 1.2 s rewriting). If you dictate in quick bursts, 1 min – 1 h keeps them loaded in between, at 0 % CPU.

## Where the memory goes

| While a dictation is processed        | Memory  | Notes                                                                                                                                                                                                 |
|---------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rewriting model weights (Qwen 3.5 2B) | ~0.9 GB | Memory-mapped straight from the model file: shared, file-backed pages that Windows can drop at any time. They stay in the file cache after the engine exits, which is why reloading takes only 0.5 s. |
| Rewriting engine, private             | ~0.5 GB | Compute buffers, attention and recurrent state, the ~27 MB prompt snapshot, the 248k-token vocabulary.                                                                                                |
| Speech engine (Whisper Small Q8)      | ~0.4 GB | Model weights (264 MB) plus encoder/decoder buffers.                                                                                                                                                  |

Task Manager's default *Memory* column counts only private memory, so it shows less than Audian's own resource card, which shows each engine's full working set. Version 0.2.0 peaked at 1.96 GB for the rewriting engine: llama.cpp kept a second, "repacked" copy of most weights (0.5 GB) and reserved a 0.5 GB compute buffer for 512-token batches. Audian now uses the mapped weights directly and 128-token batches, with no measurable change in rewriting speed.

## Only working when necessary



- The microphone is opened only while recording (the Windows mic indicator is off otherwise).
- The engines are separate processes started by the shortcut, not at boot, and they exit as soon as the dictation is done — also when it is cancelled or contained no speech — or after the idle time you chose. A Windows Job Object guarantees they never outlive Audian.
- The tray app uses no timers while idle: key-release polling runs only while you hold the shortcut, voice detection only during hands-free dictation, and the overlay animation only while it is visible.
- Preview transcriptions happen only during pauses in hands-free mode.
- OpenMP is disabled in the rewriting engine, so its threads don't spin-wait after each request.
- The settings window is a separate process, so its GUI memory exists only while it is open.
- After each dictation the tray app returns its temporary memory to Windows.

## 9 Models: what we chose and why

### Speech recognition — two engines

| Engine                                          | Where it runs                       | RAM while working    | Languages                           | Choose it when...                                         |
|-------------------------------------------------|-------------------------------------|----------------------|-------------------------------------|-----------------------------------------------------------|
| <b>NVIDIA Parakeet TDT 0.6B v3</b><br>(default) | Your CPU, offline<br>(ONNX Runtime) | ~0.8 GB              | 25 European, detected automatically | you speak one of its languages: fastest and most accurate |
| <b>Whisper</b>                                  | Your CPU, offline                   | 0.25–0.8 GB by model | 99, custom vocabulary               | you speak a non-European language or need the glossary    |

**Parakeet** is a 600M-parameter FastConformer with a Token-and-Duration Transducer decoder: it predicts each word piece together with how many audio frames to skip, so it is non-autoregressive in practice and very fast. On this PC it transcribes 15 s of speech in about 0.5 s (Whisper Small: about 1.4 s), adds punctuation by itself and rarely invents text during silence. It covers 25 European languages (English, Portuguese, Spanish, French, German, Italian, Dutch, Polish, Russian, Ukrainian and more). Audian runs NVIDIA's model (CC-BY-4.0) from an int8 ONNX export (670 MB, downloaded by the installer) in its own engine process, like Whisper. It became the default in 1.2.0.

### Whisper models

Whisper (OpenAI, MIT licence) running through whisper.cpp is the most mature offline speech engine on Windows, handles 99 languages including Portuguese, and needs no GPU. On a CPU the decisive factor is speed:

| Model                     | Disk   | RAM     | Speed on this PC | Portuguese error rate* | Choose it when...                         |
|---------------------------|--------|---------|------------------|------------------------|-------------------------------------------|
| Whisper Base              | 148 MB | ~250 MB | ≈0.2 s           | ~13.9 %                | old or low-power PCs                      |
| <b>Whisper Small Q8</b>   | 264 MB | 456 MB  | 0.4–1.4 s        | ~7.7 %                 | best Whisper speed/accuracy on CPU        |
| Whisper Small English     | 264 MB | 456 MB  | ≈0.4 s           | —                      | you only speak English                    |
| Whisper Medium Q5         | 539 MB | ~800 MB | ≈2–4 s           | ~5.1 %                 | middle ground                             |
| Whisper Large v3 Turbo Q5 | 574 MB | 777 MB  | 3–8 s            | ~4.2 %                 | accents, noisy audio; or with a GPU build |

\* Word error rate on the FLEURS Portuguese test set (public benchmark at Q8, lower is better).

**Why Small:** it is 4–6× faster than Large v3 Turbo on this CPU while being good enough for clean dictation — and the rewriting step fixes most remaining recognition slips (“end” → “and”). Turbo is the better choice once speed is not limiting (e.g. a GPU build). For English and Portuguese, Parakeet is the faster and more accurate local choice and therefore the default; Whisper remains for every other language and for the custom vocabulary.

## Rewriting — Qwen 3.5 2B (Q4\_K\_M) by default

| Model                                         | Disk         | RAM          | Rewrite time*                         | Strengths / limits                                                                                                                                                                      |
|-----------------------------------------------|--------------|--------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qwen 3.5 0.8B Q4<br>( <i>Fast</i> )           | 533 MB       | ~0.7 GB      | 0.3–0.6 s                             | About 1.5× faster on the CPU (the 248k-word vocabulary costs the same in both); fine for removing fillers, but misses most self-corrections and is weaker at restructuring (AI Prompt). |
| <b>Qwen 3.5 2B Q4</b><br>( <i>Balanced</i> )  | 1.28 GB      | 1.4 GB       | 0.5–1.5 s (≈0.1 s on a graphics card) | <b>Default.</b> Strong instruction following for its size, 100+ languages with good Portuguese, Apache-2.0.                                                                             |
| Qwen 3.5 4B Q4                                | 2.74 GB      | ~2.9 GB      | ≈1.5–2.5 s                            | Better on long, rambling dictation; needs 16 GB RAM.                                                                                                                                    |
| Llama 3.2 3B Instruct Q4                      | 2.02 GB      | ~2.5 GB      | ≈1–2 s                                | Good English writing; weaker in Portuguese.                                                                                                                                             |
| Gemma 4 E2B Q4<br>(not in catalog)            | 3.1 GB       | ~3.4 GB      | ≈1.5 s                                | Strong multilingual; large file for its speed.                                                                                                                                          |
| SmolLM3 3B / Phi-4-mini 3.8B (not in catalog) | 1.9 / 2.5 GB | 2.3 / 2.9 GB | ≈1–2 s                                | Good English; fewer languages.                                                                                                                                                          |
| Antigravity (cloud)                           | 0            | not measured | 4–8 s                                 | Best quality; needs internet; text leaves the PC.                                                                                                                                       |

\* Typical sentence on this CPU. RAM is the engine's working set while rewriting, including the memory-mapped model file; figures with “~” are estimates from file size and context shape.

**Why Qwen 3.5 2B:** rewriting needs a model that follows rules precisely (keep the meaning, don't answer questions, keep the language) while being fast enough to feel instant. In our tests the 2B model resolved self-corrections, removed fillers, kept Portuguese as Portuguese, never answered questions and translated reliably, at under a second and a half per rewrite on the CPU (about 0.1 s on a graphics card) — with about 1.3 GB of disk. The 0.8B model (*Fast*) resolved almost none of the self-corrections in our test and is weaker at AI-Prompt restructuring, so it is only worth it on a slow PC where every half second counts; 4B models are better on long inputs but 2–3× slower and need twice the RAM.

## Graphics cards

Rewriting can run on a graphics card through **Vulkan**, which NVIDIA, AMD and Intel drivers all provide, so no CUDA or extra download is needed. Audian ships two builds of the rewriting engine: `audian-llm.exe` for the CPU and `audian-llm-gpu.exe` for graphics cards (a Vulkan program cannot even start on a PC without a Vulkan driver, so the CPU build stays separate).

| Same rewrite (Qwen 3.5 2B, prompt cached) | Time             |
|-------------------------------------------|------------------|
| NVIDIA RTX 5060                           | <b>0.1–0.3 s</b> |
| Intel Core i9-12900KS, 8 threads          | 0.7–1.5 s        |
| Intel UHD 770 (built-in graphics)         | 3.4 s            |

- **Dedicated cards only.** Built-in graphics share the CPU's memory, and memory speed is what limits text generation, so they were slower than the CPU itself. Without a dedicated card Audian uses the CPU.
- **Room check.** The card needs the model's size plus 600 MB of free video memory (about 1.9 GB for the 2B model); otherwise the CPU is used, so a busy or small card never slows dictation down.
- **Fallback.** If the graphics engine crashes, hangs or cannot load, the dictation is retried on the CPU and the CPU is used until Audian restarts. *Rewriting › Use the graphics card* turns it off for good.
- **First use.** The driver compiles its compute shaders the first time (about 6 s on the RTX 5060) and keeps them in its own cache. Audian does this once in the background after installing or changing the model, together with preparing the prompt, so your first dictation is not delayed.
- **Memory.** On the card the weights are read into video memory instead of being memory-mapped, so the engine uses about 0.6 GB of RAM and 1.4 GB of video memory while working, and nothing once it exits.

Any whisper.cpp `ggml-*.bin` model or instruction-tuned GGUF chat model placed in the models folder appears in the app as a "custom" model.

# 10 How it works inside

## Process model

| Process                             | Role                                                                         | Technology                                                                   |
|-------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <code>audian.exe</code> (tray)      | Hotkeys, recording, voice detection, overlay, tray, insertion, orchestration | Rust; raw Win32 message loop; WASAPI via cpal; tiny-skia + fontdue rendering |
| <code>audian.exe -- settings</code> | The Audian window                                                            | egui (OpenGL), separate short-lived process                                  |
| <code>audian-stt.exe</code>         | Speech recognition (Whisper)                                                 | whisper.cpp (ggml, AVX2)                                                     |
| <code>audian-parakeet.exe</code>    | Speech recognition (Parakeet)                                                | ONNX Runtime 1.28, statically linked; greedy TDT decoding                    |
| <code>audian-llm.exe</code>         | Local rewriting on the CPU                                                   | llama.cpp (ggml, AVX2, no OpenMP)                                            |
| <code>audian-llm-gpu.exe</code>     | Local rewriting on a graphics card                                           | The same engine built with llama.cpp's Vulkan backend                        |
| <code>agy.exe</code> (optional)     | Cloud rewriting through Antigravity                                          | Antigravity CLI with a tool-less custom agent                                |

Engines talk to the tray app over their stdin/stdout pipes with a framed protocol (length-prefixed JSON control messages, binary frames for 16 kHz PCM audio). Separate processes keep the tray app tiny, return model memory to Windows on exit, isolate crashes in native code, and let whisper.cpp and llama.cpp each use their own ggml library (they cannot be linked into one binary).

## A dictation, step by step

1. **Shortcut** ( `RegisterHotKey` ; conflicts are reported). Audian notes the focused window, its caret position, process and title (title is used only for per-app modes, never logged).
2. **Recording** starts within ~20 ms (WASAPI shared mode, device's native rate). In parallel, the engines start and load their models; the rewriting engine restores its prompt prefix from the cache.
3. **Voice detection** on 10 ms blocks; the overlay animates from the same levels.
4. **End**: key release (push-to-talk), second tap, or auto-stop.
5. **Audio preparation**: windowed-sinc resampling to 16 kHz, trimming of leading/trailing silence, normalisation.
6. **Transcription** (unless the pause preview can be reused).
7. **Rewriting** with the session's mode (unless it was already done during the final pause); output sanitised (strip echoed tags/labels/quotes, reject outputs far longer than the input — the sign of an answer instead of a rewrite).
8. **Insertion**: verify/restore focus → wait for modifier keys to be released → clipboard snapshot → paste → restore clipboard after 0.6 s unless you copied something new meanwhile.
9. **History & stats** recorded (text only); engines exit (with “After use”); memory trimmed.

## Voice activity detection

Each 10 ms block's loudness (dBFS) is compared with an adaptive noise floor. The floor follows the quietest recent level — falling quickly, rising only 3 dB/s — so steady noise like fans is learned within seconds while the natural dips between words keep it from creeping up during speech. A block is voiced when it is *margin* dB above the floor (16 dB at sensitivity 0, 6 dB at sensitivity 1; default 11 dB; never below −60 dBFS). Speech starts after 3 voiced blocks (30 ms: ignores clicks) and ends after 200 ms of unvoiced blocks (bridges gaps between words). The detector also rejects recordings with no speech before any AI runs.

## Speech recognition optimisations

| Technique                                                                | Effect (measured)                                                            |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Compiler optimisation + AVX2 forced for whisper.cpp under MSVC           | 21.4 s → 2.4 s for a 6.6 s clip (the default MSVC build had no optimisation) |
| Encoder context sized to the audio length instead of a fixed 30 s window | 2.4 s → 0.37 s                                                               |
| Warm-up inference while you speak                                        | removes a ~1.5 s first-dictation penalty                                     |
| Language detection restricted to your languages                          | avoids misdetection on short phrases                                         |
| Glossary prompt from your custom vocabulary                              | favours your spellings of names and jargon                                   |

## Local rewriting

- **Prompt:** strict rules (edit, never answer; keep language, meaning, names; remove fillers; resolve self-corrections) + a mode style + your custom instructions + 11 few-shot examples written like real speech-recognition output, in English and Portuguese (5 of them self-corrections).
- **Self-correction hints:** passages that look like a correction are quoted after the transcript in a `<self-correction>` tag (see chapter 5). Naming the correction phrase itself was tried and made results worse: the model then deleted just the phrase and kept both versions.
- **Prefix caching:** the system prompt and examples (~1 200 tokens) are evaluated once and the model state is snapshotted (~34 MB), in memory and on disk ( `%LOCALAPPDATA%\Audian\cache` , the 6 most recent prompts, separately for the CPU and the graphics card). A freshly started engine restores it in ~25 ms instead of re-evaluating it (~6 s on the CPU, 0.3 s on the RTX 5060), and each dictation only processes its own ~40–60 tokens. Qwen 3.5 is a hybrid recurrent model, so a state snapshot is used rather than truncating a KV cache.
- **Lean memory:** weights are used straight from the memory-mapped file (llama.cpp's repacked weight copy is disabled) and the micro-batch is 128 tokens, which keeps the compute buffer — sized by the vocabulary's logits — at ~0.13 GB instead of ~0.5 GB.
- **Decoding:** greedy (deterministic), “thinking” disabled, output length capped relative to the input.
- **Translation** uses example answers in the target language, which a 2B model follows far more reliably than an instruction alone.
- **Rewriting during pauses:** in hands-free mode, when the pause preview sounds like a finished sentence, the local model rewrites it straight away while Audian waits to see whether you go on. If you don't, that text is inserted with no further wait; if you do, it is discarded. This is never done with the cloud provider.

## Antigravity integration

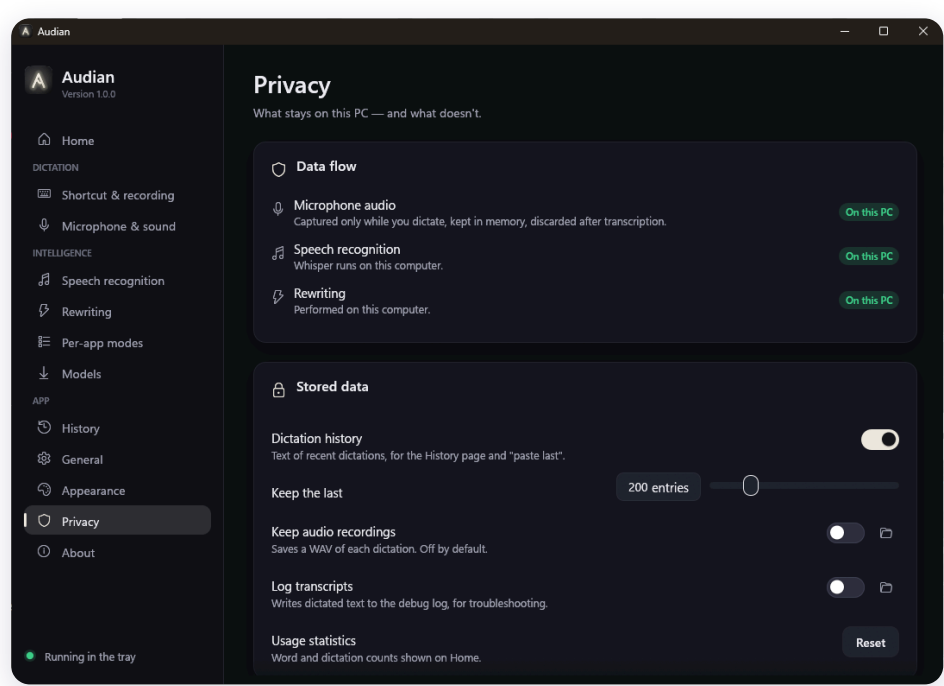
Audian keeps a private workspace ( `%LOCALAPPDATA%\Audian\agy-workspace` ) containing a custom agent with no tools and a minimal prompt — about 700 input tokens per request instead of ~12 000 for the default coding agent. The CLI is started in streaming mode when you press the shortcut, so its start-up overlaps your speech.

## Text insertion details

- Clipboard snapshot copies every memory-based format (text, rich text, HTML, images as DIB, file lists), up to 64 MB.
- Dictated text is tagged `ExcludeClipboardContentFromMonitorProcessing` , `CanIncludeInClipboardHistory=0` , `CanUploadToCloudClipboard=0` .
- For Alt/Win shortcuts, an unassigned key is injected so releasing the modifier doesn't open menus or Start.
- Elevation is detected by comparing token integrity levels; elevated targets receive the text via clipboard + notification.
- "Simulate typing" sends Unicode key events (newlines as Shift+Enter so chat apps don't send early); used as fallback when the clipboard is locked.

# 11 Privacy & your data

| Data                    | Where it goes                                        | Stored?                                                                                                    |
|-------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Microphone audio        | Memory of the tray app → speech engine (local pipe). | No. Optional “Keep audio recordings” saves WAV files (off by default).                                     |
| Transcript & final text | Local engines; or Antigravity if you selected it     | History (text only, last 200, can be turned off/cleared). Not logged unless you enable transcript logging. |
| Window title            | Used in memory to match per-app modes                | Never stored or logged.                                                                                    |
| Statistics              | Counters only (dictations, words, seconds)           | <code>stats.json</code> ; resettable.                                                                      |



The Privacy page shows the data flow for your current settings.

## Files

| What                | Location                                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Program             | <code>%LOCALAPPDATA%\Programs\Audian</code>                                                                                                   |
| Settings            | <code>%APPDATA%\Audian\config.toml</code>                                                                                                     |
| Models              | <code>%LOCALAPPDATA%\Audian\models</code>                                                                                                     |
| History, statistics | <code>%LOCALAPPDATA%\Audian\history.jsonl</code> , <code>stats.json</code>                                                                    |
| Logs                | <code>%LOCALAPPDATA%\Audian\logs</code> (rotated at 2 MB)                                                                                     |
| Prompt cache        | <code>%LOCALAPPDATA%\Audian\cache</code> — the rewriting engine's evaluated instructions and examples (never your dictations); safe to delete |

# 12 Troubleshooting

| Symptom                                                                   | What to do                                                                                                                                                                                                        |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notification “Shortcut conflict”                                          | Another app owns that shortcut. Pick another under Shortcut & recording (the editor checks availability live).                                                                                                    |
| “No speech” after speaking                                                | Check the microphone on Microphone & sound (Test microphone). Raise input gain or voice-detection sensitivity, or lower the silence threshold (the yellow mark on the meter).                                     |
| Stops before I'm finished (hands-free)                                    | Choose <i>Patient</i> or raise the pause tolerance; keep smart detection on.                                                                                                                                      |
| Text copied instead of inserted                                           | The target was closed or runs as administrator. Press <b>Ctrl</b> + <b>V</b> . To dictate into admin apps, run Audian as administrator too.                                                                       |
| Nothing pastes in a specific app                                          | General › Method › Simulate typing.                                                                                                                                                                               |
| One slower dictation after changing mode, language or custom instructions | Normal: the rewriting engine prepares the new prompt once (~6 s on the CPU, well under a second on a graphics card) and caches it; later dictations restore it in milliseconds.                                   |
| Rewriting is slow on a PC without a graphics card                         | Use hands-free mode, where finished sentences are rewritten during the pause, so the wait mostly disappears. <i>Rewriting</i> › <i>Model</i> › <i>Fast</i> is about 1.5× faster but misses most self-corrections. |
| Problems that may come from the graphics driver                           | Turn off <i>Rewriting</i> › <i>Use the graphics card</i> ; the CPU is then always used. Updating the graphics driver usually helps.                                                                               |
| Back-to-back dictations feel slow on an older PC                          | Set “Keep the model in memory” to 1–5 minutes so the engines stay loaded between dictations.                                                                                                                      |
| Wrong language recognised                                                 | Speech recognition › Languages you speak, or fix the spoken language.                                                                                                                                             |
| Antigravity errors                                                        | Make sure Antigravity is installed and signed in; the notification names the cause (sign-in, quota, network, model). Audian falls back to basic cleanup.                                                          |
| Names spelled wrong                                                       | Add them to Speech recognition › Custom vocabulary.                                                                                                                                                               |

## Diagnostics

```
audian.exe --transcribe rec.wav [--mode m] [--provider p] [--output en] > out.txt
audian.exe --rewrite-file lines.txt [--mode m] [--gpu off] ("pt|" = Portuguese line)
audian.exe --list-devices > devices.txt
audian.exe --reload (re-read config.toml after manual edits)
audian.exe --export-art art-out (logo and indicator images, every theme)
audian-stt.exe --bench ggml-small-q8_0.bin clip16k.wav 0 auto en,pt
audian-llm.exe --bench Qwen3.5-2B-Q4_K_M.gguf - "um so I want to fix the the bug"
audian-llm-gpu.exe --probe > gpu.txt (the graphics card Audian would use)
```

`audian-llm-gpu.exe` accepts the same `--bench` arguments. Set the environment variable `AUDIAN_LOG=debug` for verbose logs.

# 13 Configuration reference

`%APPDATA%\Audian\config.toml` — everything is editable in the app; manual edits are applied with *Reload configuration* (tray > More) or `audian.exe --reload`. Missing keys use defaults.

| Section · key                                                                                         | Default                                            | Meaning                                                                       |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------|
| general · start_with_windows / notifications                                                          | false / true                                       | Startup entry; error notifications                                            |
| general · sounds / sound_volume                                                                       | true / 0.5                                         | Start/stop chimes                                                             |
| hotkey · shortcut / mode                                                                              | "Ctrl+Shift+Space" / "hybrid"                      | hybrid · hold · toggle                                                        |
| hotkey · escape_cancels / paste_last_shortcut                                                         | true / "Ctrl+Shift+Alt+V"                          | Esc cancel; re-paste shortcut ("" disables)                                   |
| auto_stop · enabled / pause_secs                                                                      | true / 2.0                                         | Hands-free auto-stop and pause tolerance                                      |
| auto_stop · smart / unfinished_extra_secs / no_speech_secs                                            | true / 1.5 / 8.0                                   | Sentence-aware waiting                                                        |
| audio · device_id / device_name                                                                       | "" (system default)                                | Microphone                                                                    |
| audio · gain / vad_sensitivity / silence_threshold / max_duration_secs                                | 1.0 / 0.5 / 0.006 / 300                            | Input processing                                                              |
| transcription · provider                                                                              | "parakeet"                                         | parakeet · whisper_local                                                      |
| transcription · language / auto_languages / vocabulary                                                | "auto" / [] / []                                   | Recognition language(s), glossary                                             |
| transcription.whisper · model / threads / keep_loaded_minutes                                         | "ggml-small-q8_0.bin" / 0 / 0                      | 0 threads = auto; 0 minutes = unload right after use                          |
| transcription.parakeet · model / threads / keep_loaded_minutes                                        | "parakeet-tdt-0.6b-v3-int8" / 0 / 0                | Model folder in the models directory                                          |
| processing · enabled / mode / provider                                                                | true / "natural" / "local_llm"                     | Rewriting on/off, default mode, provider (local_llm · antigravity · rules)    |
| processing · custom_instructions / output_language / fallback_to_rules                                | "" / "same" / true                                 | Extra instructions; translation target                                        |
| processing.local_llm · model / threads / gpu / keep_loaded_minutes / max_output_tokens / timeout_secs | "Qwen3.5-2B-Q4_K_M.gguf" / 0 / true / 0 / 768 / 45 | Local rewriting engine; gpu = use a dedicated graphics card when there is one |
| processing.antigravity · executable / model / timeout_secs                                            | "" / "gemini-3.8-flash-low" / 30                   | Cloud rewriting                                                               |
| profiles · enabled / rules[]                                                                          | true / 5 rules                                     | Per-app modes (name, enabled, mode, apps, title_keywords)                     |
| insertion · method / restore_clipboard / restore_delay_ms                                             | "paste" / true / 600                               | How text is inserted                                                          |
| insertion · terminal_single_line / trailing_space                                                     | true / false                                       | Formatting                                                                    |

|                                                                          |                            |                                                                         |
|--------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------|
| overlay · position / show_label / style                                  | "auto" / true / "dark"     | auto · bottom_center · top_center;<br>dark · light                      |
| appearance · theme / mode                                                | "ivory" / "dark"           | ivory · violet · ocean · mint · sunset<br>· rose; dark · light · system |
| privacy · history / history_limit / save_recordings<br>/ log_transcripts | true / 200 / false / false | Stored data                                                             |

## 14 Building & publishing

Requirements: Rust (MSVC toolchain), Visual Studio 2022 Build Tools with *C++ build tools, Windows 11 SDK* and *C++ Clang Compiler for Windows*, plus CMake and the [Vulkan SDK](#) (for `audian-llm-gpu`, which `build.ps1` builds in a separate step and a short folder, as its shader generator would otherwise exceed the compiler's 260-character path limit).

```
.\scripts\build.ps1      # app and engines → target\release
.\scripts\package.ps1   # + installer      → dist\Audian-Setup-1.2.0-x64.exe (prints size and SHA-256)
cargo test --release     # unit tests (config, IPC, VAD, auto-stop, hotkeys, DSP, rules, sanitizer)
```

| Crate                                     | Role                                                                |
|-------------------------------------------|---------------------------------------------------------------------|
| audian                                    | Tray app + Audian window                                            |
| audian-stt / audian-parakeet / audian-llm | Speech (Whisper, Parakeet) and rewriting engines                    |
| audian-llm-gpu                            | The rewriting engine built with Vulkan (same source as audian-llm)  |
| audian-setup                              | Installer/uninstaller; embeds the app binaries (deflate-compressed) |
| audian-common                             | Config, model catalog, downloader, history, IPC, logging, WAV       |
| audian-ui / audian-art / audian-build     | Design system · vector artwork · icon/manifest/version embedding    |

### Before publishing

The installer is not code-signed, so Windows SmartScreen will warn users downloading it ("Windows protected your PC"). Signing it with a code-signing certificate (or distributing through the Microsoft Store / winget) removes the warning. Models are downloaded from Hugging Face on the user's machine — check the model licences (Parakeet: CC-BY-4.0; Whisper: MIT; Qwen 3.5: Apache-2.0; Llama 3.2: Llama Community License) if you redistribute them.